WORKED SOLUTION

STEP III 2001, Question 13

Probability and Statistics • Coin Sequence Games

STEP III 2001, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.8, *Penney's Game*: https://beyondstep.keithmcnulty.org/ch12/penney.html

The key idea

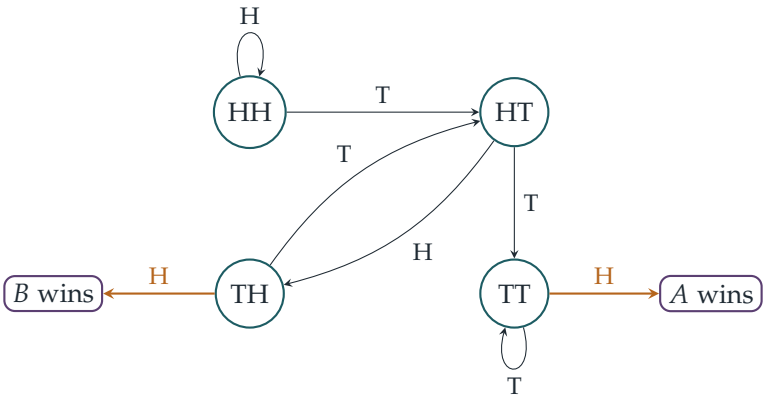
Both target sequences have length three, so nobody can win before the third toss, and from then on all that matters for the future is the *last two tosses*. That gives just four "states" (HH, HT, TH, TT), and conditioning on the next toss turns the game into a small system of linear equations in the winning probabilities from each state. Parts (ii) and (iii) can be done the same way, though in two of the four games a short direct argument is quicker.

Solution

Throughout, the tosses are independent and each shows H or T with probability $\frac{1}{2}$. Once at least two tosses have been made and nobody has won yet, the next toss together with the last two determines whether someone wins now, and if not, the new last two tosses. Earlier tosses are irrelevant, because each target sequence is only three long. So the probability of winning from a given point depends only on the last two tosses, and it is the same as if those two had been the first two tosses of a new game.

Part (i)

Let p_{XY} be the probability that A (TTH) beats B (THH) given that the first two tosses are XY. The diagram shows how the next toss moves us between states.



Why $p_{\mathrm{TT}} = 1$. Suppose the last two tosses are TT. With probability 1 a head eventually appears (the chance of n tails in a row is $2^{-n} \to 0$). Until it does, the tosses are all tails, so THH cannot appear; and the first head is preceded by at least two tails, so it completes TTH. Hence A wins for certain: $p_{\mathrm{TT}} = 1$.

Why $p_{\mathrm{HT}} = \frac{1}{2}p_{\mathrm{TH}} + \frac{1}{2}p_{\mathrm{TT}}$. After HT, neither HTH nor HTT is a target sequence, so nobody wins on the third toss. If it is H (probability $\frac{1}{2}$) the last two tosses are TH; if it is T they are TT. Conditioning on the third toss, and using the remark above that only the last two tosses matter,

$$p_{\mathrm{HT}} = \tfrac{1}{2}p_{\mathrm{TH}} + \tfrac{1}{2}p_{\mathrm{TT}}.$$

The other two states. After TH, a head gives THH and B wins, while a tail leaves HT. After HH, a head leaves HH and a tail leaves HT (and HHH, HHT are not targets here). So

$$p_{\mathrm{TH}} = \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}p_{\mathrm{HT}}, \qquad p_{\mathrm{HH}} = \tfrac{1}{2}p_{\mathrm{HH}} + \tfrac{1}{2}p_{\mathrm{HT}}.$$

The second equation gives $p_{\mathrm{HH}} = p_{\mathrm{HT}}$. Substituting $p_{\mathrm{TH}} = \frac{1}{2}p_{\mathrm{HT}}$ and $p_{\mathrm{TT}} = 1$ into the equation for p_{HT},

$$p_{\mathrm{HT}} = \tfrac{1}{4}p_{\mathrm{HT}} + \tfrac{1}{2} \quad\Longrightarrow\quad \tfrac{3}{4}p_{\mathrm{HT}} = \tfrac{1}{2} \quad\Longrightarrow\quad p_{\mathrm{HT}} = \tfrac{2}{3}.$$

Hence $p_{\mathrm{HH}} = p_{\mathrm{HT}} = \frac{2}{3}$ and $p_{\mathrm{TH}} = \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{3}$, as required.

Ask yourself Why is it legitimate to "restart" the game from TH, as if TH were the first two tosses, when TH actually arose at tosses 5 and 6, say?

Because the future of the game depends only on future tosses and the last two tosses: any occurrence of TTH or THH that is not yet complete must use at most the last two tosses already made. The earlier history cannot help either sequence, and the future tosses are independent of it.

The overall probability. Nobody can win in the first two tosses, and the four possible pairs each have probability $\frac{1}{4}$. By the law of total probability,

$$\mathrm{P}(A \text{ wins}) = \tfrac{1}{4}\left(\tfrac{2}{3} + \tfrac{2}{3} + \tfrac{1}{3} + 1\right) = \tfrac{1}{4}\cdot\tfrac{8}{3} = \tfrac{2}{3}.$$

Part (ii)

Now B has THH and C has HHT. Here a direct argument is neatest.

If the first two tosses are HH, then C wins: the first tail after them (which comes with probability 1) is preceded by HH and completes HHT, and before it every toss is a head, so THH cannot have appeared.

Otherwise there is a tail among the first two tosses. With probability 1 the pattern HH eventually occurs. Consider its first occurrence. It does not start at toss 1, so the toss just before it exists, and it is not H (else there would be an earlier HH), so it is T. Thus the first HH completes a THH. Meanwhile HHT cannot have occurred before, since it would need an earlier HH. So B wins.

Therefore B wins unless the first two tosses are HH:

$$\mathrm{P}(B \text{ wins}) = 1 - \tfrac{1}{4} = \tfrac{3}{4}.$$

Part (iii)

C (HHT) against D (HTT). Let q_{XY} be the probability that C wins from last two tosses XY. From HH, C wins for certain, by the same reasoning as in (ii): the first tail completes HHT, and HTT needs a tail after HT, which cannot come first. So $q_{\mathrm{HH}} = 1$. Conditioning on the next toss in each other state:

HT:	T gives HTT (D wins), H gives TH:	$q_{\mathrm{HT}} = \frac{1}{2} q_{\mathrm{TH}}$,
TH:	H gives HH, T gives HT:	$q_{\mathrm{TH}} = \frac{1}{2} + \frac{1}{2} q_{\mathrm{HT}}$,
TT:	H gives TH, T gives TT:	$q_{\mathrm{TT}} = \frac{1}{2} q_{\mathrm{TH}} + \frac{1}{2} q_{\mathrm{TT}}$.

The last gives $q_{\mathrm{TT}} = q_{\mathrm{TH}}$. Substituting the first into the second, $q_{\mathrm{TH}} = \frac{1}{2} + \frac{1}{4} q_{\mathrm{TH}}$, so $q_{\mathrm{TH}} = \frac{2}{3}$, $q_{\mathrm{TT}} = \frac{2}{3}$ and $q_{\mathrm{HT}} = \frac{1}{3}$. Hence

$$\mathrm{P}(C \text{ wins}) = \tfrac{1}{4}\left(1 + \tfrac{1}{3} + \tfrac{2}{3} + \tfrac{2}{3}\right) = \tfrac{1}{4} \cdot \tfrac{8}{3} = \tfrac{2}{3} > \tfrac{1}{2},$$

so C is more likely to win than D.

D (HTT) against A (TTH). This mirrors part (ii) with H and T swapped. If the first two tosses are TT, A wins: the first head completes TTH, and HTT would need a head followed by two tails, which cannot happen before that first head. Otherwise there is a head among the first two tosses. With probability 1 the pattern TT eventually occurs; at its first occurrence it does not start at toss 1, and the toss before it cannot be T (there would be an earlier TT), so it is H and the first TT completes HTT. TTH cannot have appeared earlier, as it needs an earlier TT. So D wins, and

$$\mathrm{P}(D \text{ wins}) = 1 - \tfrac{1}{4} = \tfrac{3}{4} > \tfrac{1}{2}.$$

So D is more likely to win than A.

? **Ask yourself** A beats B, B beats C, C beats D and D beats A. How can that be?

"Is more likely to beat" is not a transitive relation, so there is no contradiction: the four players form a cycle, with A, B, C, D winning with probabilities $\frac{2}{3}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{3}{4}$ in their respective games. Whoever chooses second can always pick a sequence that beats the first player's.

Checks

- In part (i), substitute back: $\frac{1}{2}\cdot\frac{1}{3}+\frac{1}{2}\cdot 1=\frac{2}{3}=p_{\mathrm{HT}}$, and $\frac{1}{2}\cdot\frac{2}{3}=\frac{1}{3}=p_{\mathrm{TH}}$.
- The state method also works for part (ii) and the D–A game: for B against C the winning probabilities for B from HH, HT, TH, TT are $0, 1, 1, 1$, averaging to $\frac{3}{4}$.
- A simulation of 200 000 games for each pairing gave winning frequencies 0.664, 0.752, 0.666 and 0.749, against the exact $\frac{2}{3}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{3}{4}$.

Common slips

- Forgetting that the game can end on the third toss, for example writing $p_{\mathrm{TH}}=\frac{1}{2}p_{\mathrm{HH}}+\frac{1}{2}p_{\mathrm{HT}}$ instead of recognising that TH followed by H is an immediate win for B.
- Averaging the four state probabilities without saying that each pair of first tosses has probability $\frac{1}{4}$ and that nobody can win before the third toss.

Beyond STEP

This is Penney's game, a famous example of non-transitive probabilities. The state method is the theory of absorbing Markov chains, where the probabilities of absorption satisfy exactly the kind of linear equations solved here. There is even a closed formula for the odds between any two sequences, Conway's "leading numbers" algorithm, which can be proved using martingales. For Markov chains and first-step analysis, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

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